

Term Report (ESET 369)

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Abstract - This report provides an overview of the key topics and laboratories completed in ESET 369, and presents an embedded system design utilizing these concepts.

I. Introduction

Embedded systems are fundamental in everyday electronics, Used for integrating software and hardware to perform tasks. This report highlights major concepts used to design and implement systems such as GPIO, timers, interrupts, ADC, PWM, serial communication, UART, SPI, I2C, and other related topics.

Additionally these concepts are applied to demonstrate how they can be implemented to create the functionality of a vault/safe with a servo to move the open and close the door, RFID as the key to open the vault, and a LCD to display feedback to the user.

II. Lectures/Labs

Chapters 1 through 27 (excluding Chapters 17 and 26) of "Learning Embedded Systems with MSP430 FRAM Microcontrollers" by Byul Hur were covered in lecture. The textbook begins by introducing embedded systems, the MSP430 family, and development tools like Code Composer Studio and LaunchPad kits. It explains the MSP430FR5994 architecture, focusing on its memory, clock systems, and low-power features, and transitions into programming with both assembly and C at the register level. Core topics such as GPIO control, timers, and interrupts are detailed, followed by discussions on LCD display interfacing and the analog-to-digital converter (ADC) module and its applications in sensing.

Software architecture choices are outlined, comparing round-robin, interrupt-driven, and real-time operating systems. PWM generation is introduced for motor control applications, leading into DC motor and servo motor control strategies. Serial communication fundamentals are presented, with UART, SPI, and I2C protocols explained through examples. Time measurement techniques using timers are explored, alongside introductions to wireless communication modules like Wi-Fi and Bluetooth.

System integration practices are demonstrated with educational robotics examples, followed by a description of the TI Driver Library for streamlined peripheral control. An introduction to TI-RTOS discusses multitasking fundamentals. Open-source platforms like Arduino and Energia are recommended for rapid development. The summary concludes by addressing embedded system security principles, including encryption, authentication, and secure firmware update practices.

The lab sequence for ESET 369 introduces students to embedded systems programming using the MSP430 microcontroller. Lab 1 begins by familiarizing students with the Code Composer Studio environment, basic I/O operations, and debugging skills. Lab 2 builds on this foundation by guiding students through more complex digital I/O tasks such as debouncing mechanical switches and using pull-up/pull-down resistors. Lab 3 introduces timing and delay concepts by requiring the creation of software-based delays and timed LED flashing. In Lab 4, students begin working with hardware timers and interrupt-driven programming, creating more efficient timing mechanisms than software loops. Lab 5 expands these skills into pulse width modulation (PWM), where students control LED brightness through variable duty cycles. Lab 6

introduces analog interfacing with the MSP430's analog-to-digital converter (ADC), teaching students how to read analog sensor inputs and interpret their values digitally. Lab 7 shifts focus to software structure, requiring the creation of a basic finite state machine (FSM) to organize program logic more effectively. Lab 8 applies previous knowledge into a menu-driven system design, reinforcing FSM concepts and modular programming practices. Finally, Lab 9 serves as a culmination of all skills learned by combining analog, digital, and software techniques to create a fully functional embedded system.

III. EMBEDDED SYSTEM

The embedded system was built around the MSP430FR5994 microcontroller. This system utilizes an I2C LCD module, an RFID-RC522 scanner, and a servo motor. The purpose of this conceptual embedded system is to be a control system for a secured door.

The I2C LCD module is used to interface with the user and display feedback determined through the RFID module. When a user scans their RFID tag, the system checks whether the tag is recognized. If the tag is authorized, the LCD displays a welcome message with the user's name and the servo motor is triggered to unlock the door. If the tag is not recognized, the LCD displays a denial message and the servo does not open the door.

The LCD is connected to the MSP430 using an I2C configuration with the SDA and SCL being connected to P3.0 and P3.1 respectively. The RFID-RC522 scanner is set up in a SPI configuration with MISO, MOSI, SCK, and CS being connected to P7.1, P7.0, P7.3, and P8.0 respectively. The servo is connected to an external power supply with its control being connected to P5.1.

Overall, the system integrates I2C and SPI communication protocols alongside PWM control to achieve a vault/safe functionality, combining wireless authentication, real-time user

feedback, and mechanical actuation in a compact embedded design. The full block diagram for this embedded system can be seen in **Figure 1** below.

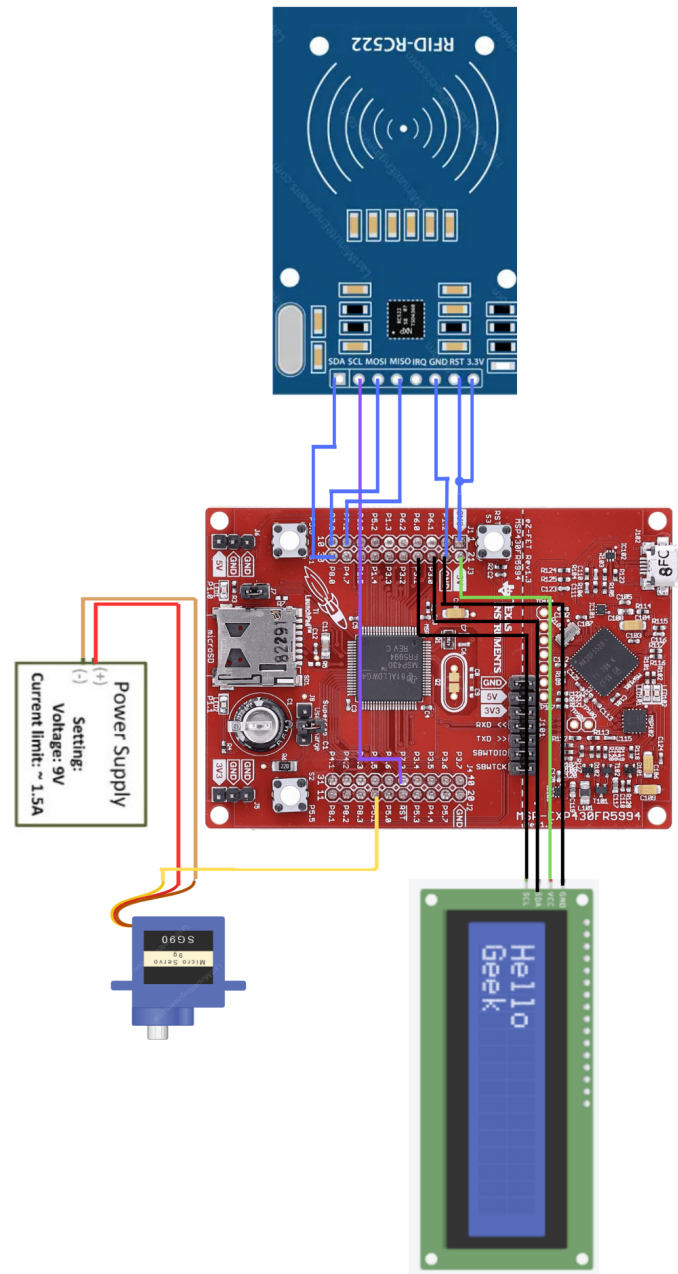


Figure 1: Embedded System Block Diagram

IV. SUMMARY AND CONCLUSION

The lecture material provided a foundational understanding of embedded systems using the MSP430 platform, emphasizing key concepts such as microcontroller architecture, memory structures, peripheral interfacing, and embedded software design. Topics including GPIO control, timers, interrupts, ADC operation, and serial communications were progressively layered to build a comprehensive technical framework. Concurrently, the lab exercises reinforced these principles through practical implementation, beginning with basic I/O and extending to PWM generation, ADC signal processing, FSM design, and system-level integration. Each lab was strategically aligned with the lectures to promote both theoretical understanding and applied proficiency in embedded system development.

Completion of the lecture and laboratory components resulted in a well-rounded skill set encompassing both low-level hardware interfacing and structured embedded software design. The cumulative approach of progressively building from simple I/O manipulation to complex system integration effectively bridged theoretical knowledge with hands-on experience. Students are now equipped with the necessary competencies to design, implement, and debug embedded systems using industry-standard tools and methodologies, positioning them for success in advanced coursework, capstone projects, or professional applications.

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